

Grade 10 - Term 3 Structure

Overview

This directory contains the source materials for Grade 10 Term 3 probability instruction. The sequence moves from practice on discrete and continuous distributions (criteria C0-C7), through student assessments and their teacher solution sets, to catch-up work and a C8-C10 web/data-modeling project. All printable resources are LaTeX source files; the only local data assets are three CSV datasets supplied for the project. There are no HTML, CSS, JavaScript, compiled PDFs, or build/configuration files in this directory.

The inventory below documents 32 existing files in 4 directories. STRUCTURE.md itself is the inventory document and is intentionally omitted from the tree.

Resource roles

- ? Student-facing pages/materials: the practice activities, context article, assessment sheets, catch-up sheet, project brief, datasets, and normal-table reference are intended for learners. In this repository they are printable LaTeX sources rather than web pages.
- ? Teacher materials: files containing solution/solutions provide answer keys or worked derivations. G10_T3_summary.tex is explicitly a pedagogical review rather than student exercises.
- ? Lesson and assessment resources: 1-practice_activities/ develops criteria C0-C7 and also contains consolidated C1-C10 material; 2-learning_evidences/ holds formal or practice assessment variants; 3-catch_ups/ provides remediation.
- ? Stylesheets and JavaScript: none are stored here. The C8-C10 brief requires students to create HTML, CSS, and JavaScript, but those deliverables are not part of this directory.
- ? Images and media: no image file is stored here. Several assessment sources request a shared logo.png outside this subtree or through the shared LaTeX search path; graphs are generated in LaTeX with TikZ/PGFPlots or shared graph macros.
- ? Data and configuration: the three CSV files are the only local datasets. There are no local configuration files.
- ? Downloadable documents: no compiled PDF or office-document download is checked in; the .tex files are sources from which printable documents can be built.
- ? Generated/build files: none are present or inventoried.

Important relationships

- ? Nearly every LaTeX document imports a shared preamble (preamble_exams.tex or preamble.tex) using relative search paths. Those shared files live outside term_3 and supply document classes, packages, layout commands, and assessment helpers; the two G10_T3_MATERIAL.tex files directly input ../../preamble/preamble_exams.tex.
- ? C4-C7 assessments and solution sets also import shared graph libraries such as graphs/probability_distribution_graphs.tex, graphs/normal_distribution_graphs.tex, and graphs/uniform_linear_probability_distributions.tex. These external macro files generate reusable uniform, triangular, and normal plots; they are not assets contained in this tree.

- ? Assessment headers in the midterm, C1-C4 exams, and catch-up sheet reference logo.png through the preamble/graphics search paths. The C8-C10 activity instead checks ../../preamble/logo.png explicitly and displays a placeholder if it is unavailable. No local copy of the logo exists.
- ? G10_T3_C1C7_midterm_solution.tex corresponds to G10_T3_C1C7_midterm.tex; their V2 counterparts form the alternate pair. V2 changes actual model bounds, means/spreads, wording, and layout-not merely the filename.
- ? G10_T3_L1_C4_exam_solution.tex and G10_T3_L1_C4_exam_solution_long.tex both answer the original C4 exam: the former is a compact grouped key, while the latter gives full per-question explanations. G10_T3_L1_C4_examV2_solution.tex is specifically paired with the numerically different V2 practice exam.
- ? G10_T3_C5C6C7_catch_upsolutions.tex is the worked companion to G10_T3_C5C6C7_catch_up.tex; both cover the same three criteria, while the solution source additionally loads shared graph macros.
- ? The C8-C10 activity instructs students to build a separate interactive webpage. Its C9 data-modeling mode is expected to accept all three adjacent CSV datasets, estimate a suitable uniform/triangular/normal model, and compare a fitted PDF with an empirical histogram. No resulting HTML, CSS, JavaScript, report, or deployed-site configuration is included here.
- ? normal-table.tex supports the standard-normal work in C5-C7 as a standalone printable reference. Some practice and solution documents embed their own shorter Phi tables, so they do not directly input this file.
- ? G10_T3_MATERIAL_V2.tex is a substantive pedagogical revision of G10_T3_MATERIAL.tex: it preserves the C1-C10 collection structure while changing early continuous-density solutions to geometry-based methods and replacing at least one nonlinear model with a piecewise-constant one.

Verification notes

- ? Every item could be identified confidently from its contents; no entry requires manual verification.
- ? No transient or generated item was present in the tree. In particular, there were no .DS_Store, editor-cache, compiler-auxiliary, temporary, dependency, compiled PDF, or LaTeX build-output files to exclude.

Directory tree: 1-practice_activities/

1-practice_activities/

Student practice, consolidated worked material, contexts, and a teacher-oriented summary for

G10_T3_C0_practice_activity.tex
Student-facing worked practice on discrete PMFs: validity, unknown probabilities, event sums, graphs, and contextual interpretations.
G10_T3_C1C2C3_practice_activity_context.tex
Student-facing contextual set joining C1 density validity, C2 uniform mixtures, and C3 triangular investment models.
G10_T3_C1_practice_activity.tex
Student-facing C1 progression from geometric area review to validating and using uniform and linear PDFs, including parameterized domains.
G10_T3_C2_practice_activity.tex
Student-facing C2 practice on uniform and piecewise-uniform PDFs, unknown constants/endpoints, complements, and multi-interval models.
G10_T3_C3_practice_activity.tex
Student-facing C3 practice using geometry-not integration-for linear, triangular, piecewise-linear, and parameterized densities.
G10_T3_C4_practice_activity.tex
Student-facing C4 worked comparisons of mean, variance, and standard deviation for uniform, triangular, and normal distributions.
G10_T3_C5_practice_activity.tex
Student-facing C5 guide and nine worked problem types for reading the standard-normal table and combining tails/intervals.
G10_T3_C6_practice_activity.tex
Student-facing C6 contextual normal problems organized by left-tail, right-tail, central-band, and two-region cases after standardization.
G10_T3_C7_practice_activity.tex
Student-facing C7 inverse-normal applications for upper/lower percentiles and symmetric central intervals in real contexts.
G10_T3_Contexts.tex
Standalone explanatory article comparing real-life continuous-PDF models, geometric calculations, assumptions, limitations, and references.
G10_T3_MATERIAL.tex
Consolidated worked C1-C10 problem collection; the original edition uses integration in early density solutions.
G10_T3_MATERIAL_V2.tex
Revised consolidated C1-C10 collection replacing early integration-based work with geometric area reasoning and revising some models.
G10_T3_summary.tex
Teacher-oriented pedagogical summary of the C0-C7 practice files, including progression, strengths, gaps, and suggested improvements.

Directory tree: 2-learning_evidences/

2-learning_evidences/

Lesson assessments, alternate versions, teacher solutions, a computational project brief, and

G10_T3_C1C7_midterm.tex
Student midterm assessing C1-C7 with original numerical parameters and contextual density/normal-distribution tasks.
G10_T3_C1C7_midtermV2.tex
Alternate student midterm with revised header/rubric, shifted numerical parameters, and reformatted C5 questions.
G10_T3_C1C7_midterm_solution.tex
Teacher solution set for the original C1-C7 midterm, with full derivations, graphs, and a minimal C5-C7 answer section.
G10_T3_C1C7_midterm_solutionV2.tex
Teacher solution set matching the V2 midterm's changed endpoints, scenarios, distribution parameters, and answers.
G10_T3_L1_C1C2C3_exam.tex
Student Learning Evidence 1 exam: 17 C1-C3 density-validity, uniform, piecewise-uniform, and linear/triangular tasks.
G10_T3_L1_C1C2C3_exam_solution.tex
Teacher worked solutions to the C1-C3 exam, including geometric derivations and plotted density regions.
G10_T3_L1_C4_exam.tex
Student C4 Learning Evidence 2 on matching/drawing uniform, triangular, and normal distributions by mean and spread.
G10_T3_L1_C4_examV2.tex
Non-credit practice version of the C4 assessment with changed distributions, graph ranges, and answer options.
G10_T3_L1_C4_examV2_solution.tex
Teacher worked answers for the altered C4 V2 practice set, organized problem by problem.
G10_T3_L1_C4_exam_solution.tex
Concise teacher answer key for the original C4 exam, grouped into matching and drawing problems and retaining the exam-style header.
G10_T3_L1_C4_exam_solution_long.tex
Expanded teacher solution for the original C4 exam with an evaluated-criterion overview and individual explanations for all 18 tasks.
G10_T3_L1_C5_exam_solution.tex
Teacher C5 worked solution set covering six standard-normal table patterns, including symmetry, complements, intervals, and tail unions.
G10_T3_L4_C8C9C10_activity.tex
Student Learning Evidence 4 brief requiring an HTML/CSS/JavaScript modeling webpage, AI-development evidence, deployment, and a statistical report.
G10_T3_L4_C8C9C10_dataset_1.csv
C9 test data: 30 measurements labeled as a uniform-like process (12.4-26.3).
G10_T3_L4_C8C9C10_dataset_2.csv
C9 test data: 30 measurements labeled as a triangular-like process (5.2-20.3), concentrated near the middle.
G10_T3_L4_C8C9C10_dataset_3.csv
C9 test data: 30 measurements labeled as a normal-like process (42.1-65.2), centered in the mid-50s.

Directory tree: 3-catch_ups/

3-catch_ups/

Remedial C5-C7 assessment material
and its teacher solution
companion.

G10_T3_C5C6C7_catch_up.tex

Student catch-up sheet on standard-normal probabilities, contextual standardization, and inverse-normal cutoffs/intervals.

G10_T3_C5C6C7_catch_upsolutions.tex

Teacher solutions for the catch-up sheet, with full calculations, distribution graphs, and a minimal-answer recap.

Directory tree: 4-extra_material/

4-extra_material/

Standalone student reference material supplementing the normal-distribution work.

normal-table.tex

Printable standard-normal reference sheet with table-reading directions, identities, a shaded curve, and Phi values through $z = 3.09$.